

SPEBT Project Update

Feb. 19 2025

Fang Han

Image Reconstruction setup

System configuration

- ▶ 24 rotations
- ▶ 864 detector units (6 panels $\times$ 144 detector units) for each rotation

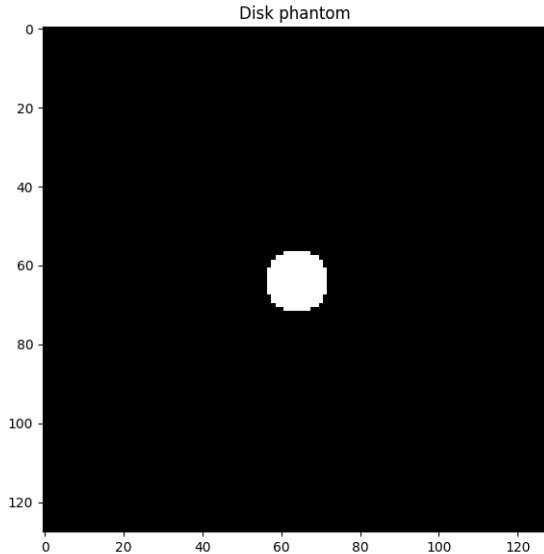
Phantoms

- ▶ disk phantom at the center
- ▶ hotrods phantom

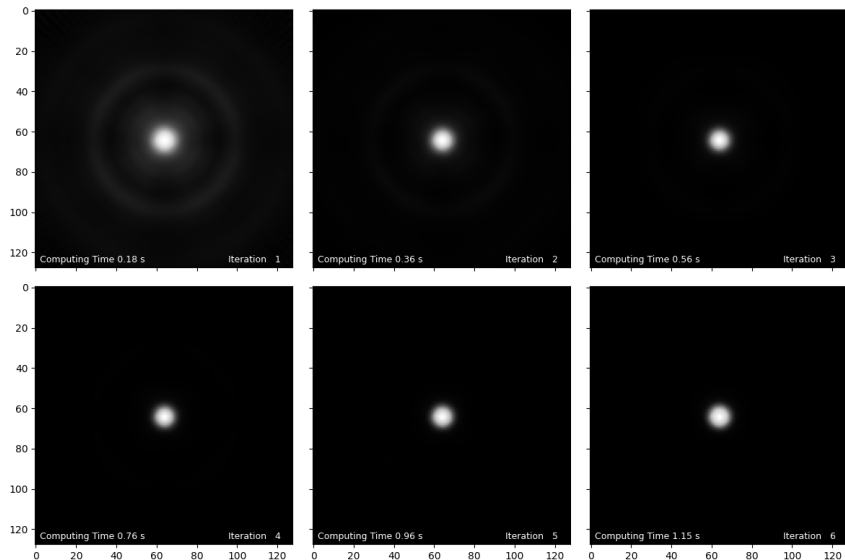
Computer

- ▶ 8 cores Intel CPU with Hyper-threading
- ▶ 16 GB memory

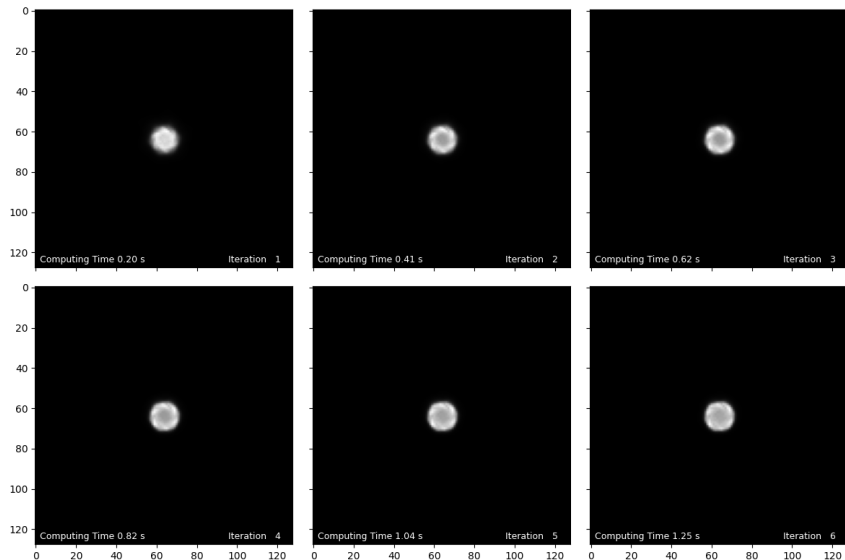
Disk phantom



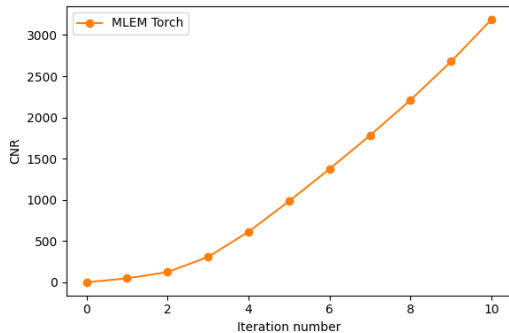
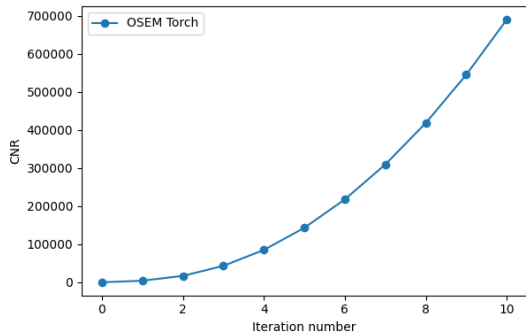
Maximum Likelihood Expectation Maximization (MLEM) Reconstruction



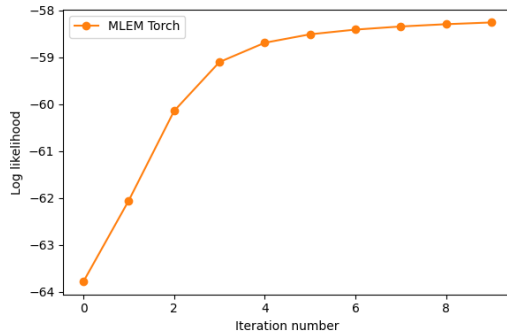
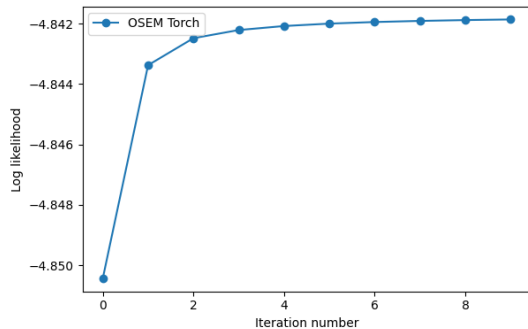
Ordered Subset Expectation Maximization (OSEM) Reconstruction



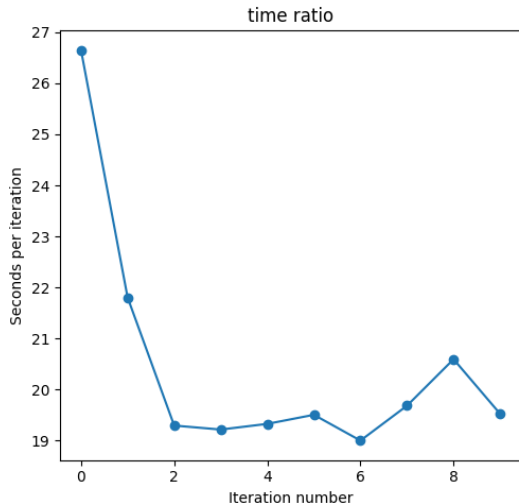
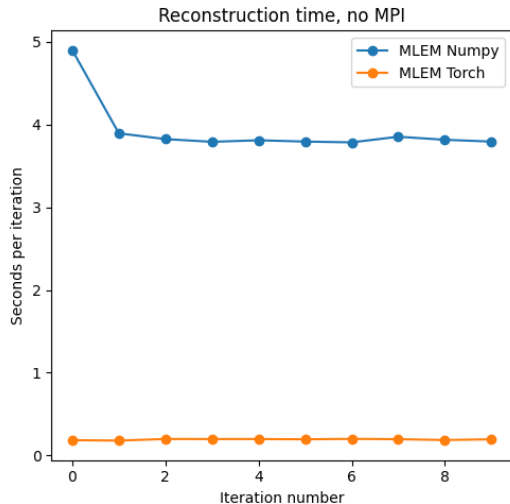
Contrast to Noise Ratio (CNR)



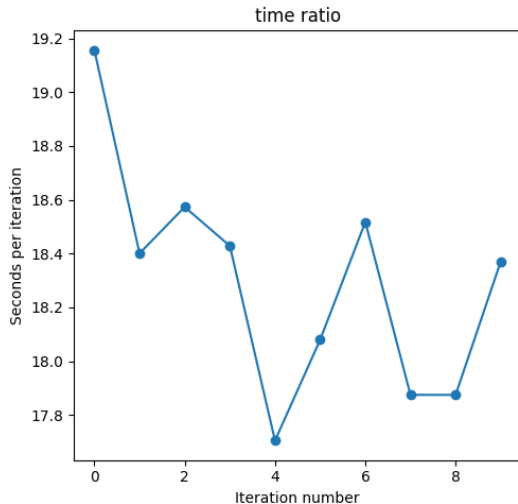
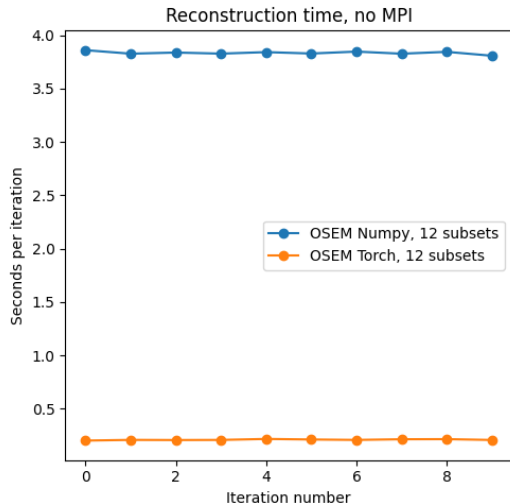
Log Likelihood



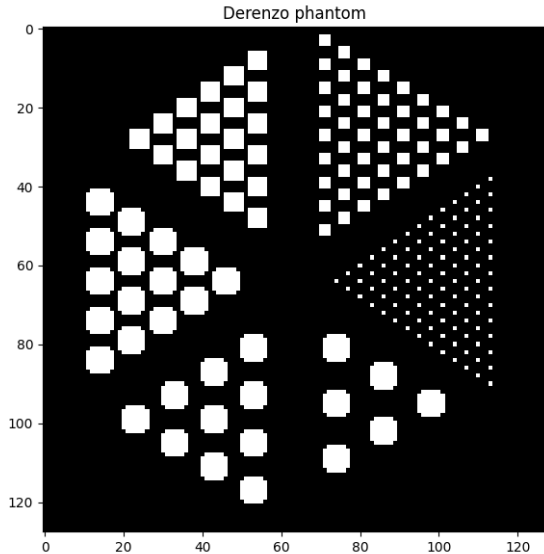
Time improvement from Numpy to PyTorch for MLEM, Disk Phantom



Time improvement from Numpy to PyTorch for OSEM, Disk Phantom

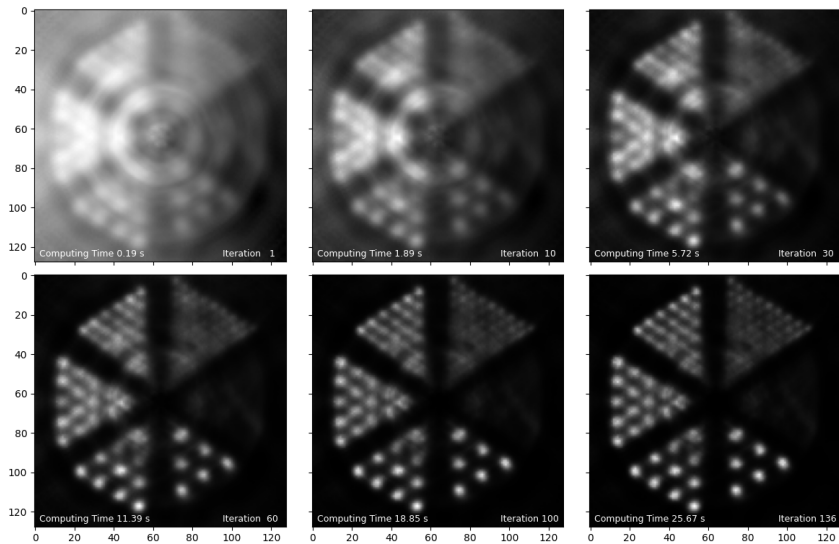


Hotrods phantom



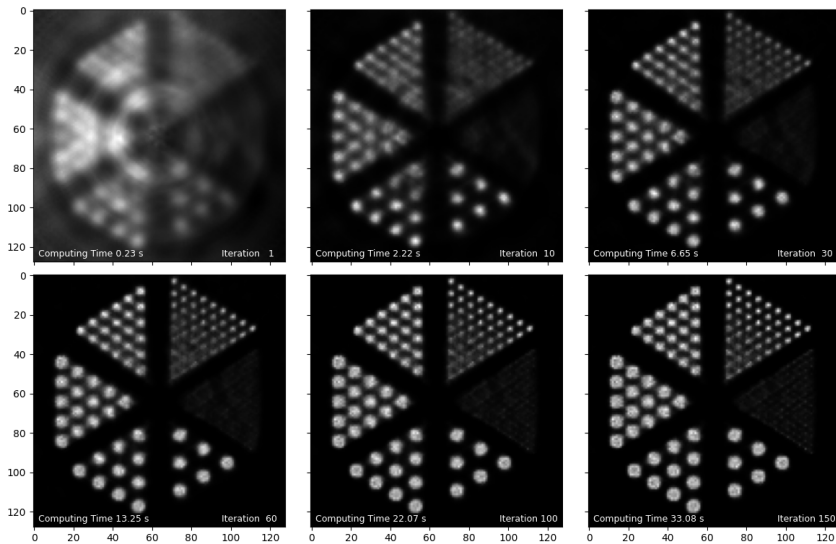
MLEM Reconstruction

MLEM Reconstruction, no MPI, PyTorch, Derenzo phantom

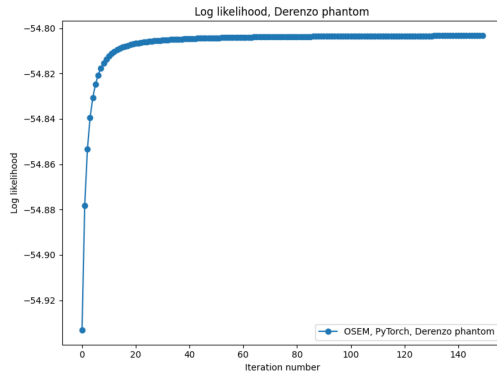
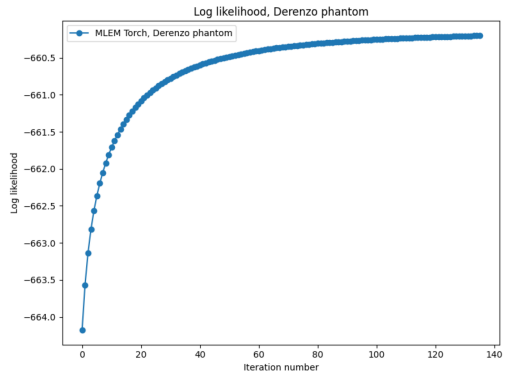


OSEM Reconstruction

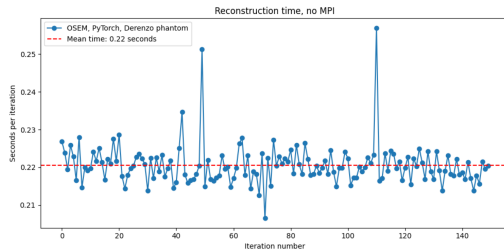
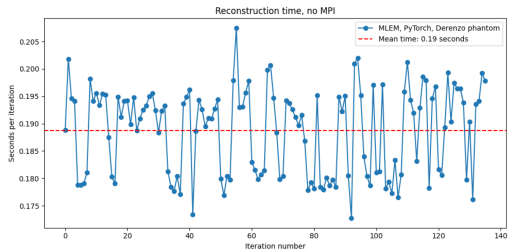
OSEM Reconstruction, no MPI, PyTorch, Derenzo phantom



Log Likelihood



Computing Time



Thanks